

Achim J. Lilienthal – CV (Summary)

GENERAL DATA

Name: Achim Josef Lilienthal
Date of Birth: July 26, 1970 (Messkirch, Germany)
Citizenship: German
Marital Status: Married to Tanja Lilienthal
Children: One daughter, Helena Luna Lilienthal
Swedish Personal Number: 700726-6591
Languages: German, English, Swedish (basic)

Tel: (+46) 19 30 3602
Fax: (+46) 19 30 3463
E-mail: achim@lilienthals.de
Web: <http://aass.oru.se/Research/Learning/amll.html>



APPOINTMENTS & FELLOWSHIPS

Jul 2005 - present
Postdoc (forskarassistent), Dept. of Technology, Örebro University.

Sep 2004 - Jun 2005
Software developer, Netzgiganten, Gerokstr. 6, 70188 Stuttgart.

Jul 2002 - Dec 2002
Marie Curie fellowship at the Learning Systems Lab, AASS, Örebro University (funded by the European Union).

Mar 1999 - Jul 2004
PhD student, WSI Computer Science Dept., University of Tübingen, Germany.

Aug 1998 - Mar 2000
Software developer, EffCom GmbH, Mühlbachstr. 20, 78351 Ludwigshafen, Germany.

PROFESSIONAL PREPARATION

PhD (Dr. rer. nat.) in Computer Science at the WSI Computer Science Department, University of Tübingen, Germany, Dec 2004.
PhD thesis: *Gas Distribution Mapping and Gas Source Localisation with a Mobile Robot*.
Grade: summa cum laude.

MSc (Diploma) in Cluster Physics at the University of Konstanz, Jun 1998.
MSc thesis: *Investigation of the Structure of $(C_{60})_+^n$ Clusters Using Gas Phase Ion Chromatography* (translated title¹).

RESEARCH ACTIVITY (SUMMARY)

2005 - now
▷ Robot Vision for Recognition Systems, including 3D Environment modelling, Object and Place Detection, Visual Person Tracking.
▷ Robotic Map Learning and SLAM (Simultaneous Localization and Mapping), including 3D Scan Registration, 3D Mapping, Semantic Mapping, and ▷ Appearance Based Topological Mapping.
▷ Safe Navigation Systems.
▷ Learning Control, in particular Programming-by-Demonstration based approaches.

1999 - now
Mobile Robot Olfaction, including gas distribution mapping and gas source localisation.

1997 - 1998
Cluster Physics.

AWARDS

Award for innovative research from TL donation (Förderpreis der TL Stiftung), 15000 Euro, March 15, 2004.

Best Paper Award, IEEE International Conference on Advanced Robotics (ICAR), June 30, 2003.

¹Original (in German): Strukturuntersuchungen an $(C_{60})_+^n$ -Clustern mit der Methode der Gasphasen-Ionenchromatographie

COMMISSIONS
OF TRUST

Services to the Scientific Community

- ▷ Program committee member of the 10th International Conference on Intelligent Autonomous Systems (IAS), Baden-Baden, Germany, July 23-25, 2008.
- ▷ Program committee member of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2007) Workshop “From sensors to human spatial concepts” (<http://staff.science.uva.nl/~zivkovic/FS2HSC/workshops/IROS2007/>).
- ▷ Joint Organization of the Workshop on Robotic Olfaction (<http://aass.oru.se/~ali/icra07ws/>) held during the IEEE International Conference on Robotics and Automation (ICRA), Rome, Italy, April 10-14, 2007.
- ▷ Member of the General Management Board and Dissemination Representative of the EU project “Dustbot – Networked and Cooperating Robots for Urban Hygiene” (FP6-2005-IST-6). (<http://www.dustbot.org>).

Referee

I am a regular reviewer for 15 international conferences and journals in the field of mobile robotics and sensors. I have been also a referee for several international workshops and an EU project.

Invited Talks and Seminars

- ▷ Invited lecture at Delft University of Technology within a symposium “Safety First: Dealing with Disasters”, Delft, Netherlands, March 13, 2008.
- ▷ Invited talk on “Vision-Based Interpolation of 3D Scans and Autonomous Difference Detection for Security Patrol Robots” at Tübingen University, GRIS, Tübingen, Germany, April 20, 2007.
- ▷ Invited talk on “The Dustbot Project and Beyond” at Halmstad Högskola, January 11, 2007.
- ▷ Invited talk on “Current Research at AASS Learning Systems Lab” within the seminar “Robots dreams, fears, and actual trends” at Urbino University, Urbino, Italy, September 10, 2005.
- ▷ Invited talk on “Gas Distribution Mapping and Gas Source Localisation with a Mobile Robot” at the Scuola Superiore Sant’Anna, Pontedera, Italy, September 8, 2005.

SUPERVISION

Ph.D. Supervision

I am and have been supervising 11 Ph.D. students. Two of them have defended their Ph.D. thesis and three have presented a Licentiate thesis under my supervision (the Licentiate is a thesis Ph.D. candidates can write in Sweden after half the time of their Ph.D. studies). I further have been supervising three M.Sc. and two B.Sc. projects.

TEACHING

Postgraduate Courses

Machine Learning 2007 (together with Thorsteinn Rögnvaldsson); lectures, seminars (42/2 h), course development.

Undergraduate Courses

- ▷ Digital Image Processing 2007; lectures (28 h), examination.
- ▷ Digital Image Processing 2006; lectures (28 h), examination, course / lab assignment development.
- ▷ Web Development with J2EE 2006/07; lectures (16 h), examination.
- ▷ Web Development with J2EE 2005/06; lectures (16 h), examination, course / lab assignment development.
- ▷ Web Client Programming 2006/07; lectures (16 h), examination, course / lab assignment development. Substitutional lecturers and lab assistance in Robotics and Practical Physics (in Germany).

Achim J. Lilienthal: